Electrodynamics and Relativity

Problem 12.1

Problem 12.2

Problem 12.3

Problem 12.4

As the outlaws escape in their getaway car, which goes $\frac{3}{4}c$, the police officer fires a bullet from the pursuit car, which only goes $\frac{1}{2}c$. The muzzle velocity of the bullet (relative to the gun) is $\frac{1}{3}c$. Does the bullet reach its target (a) according to Galileo, (b) according to Einstein?

Solution:

(a) Adding up the velocities with Galileo's velocity addition rule, the resulting velocity of the bullet is

$$\frac{1}{2}c + \frac{1}{3}c = \frac{5}{6}c > \frac{3}{4}c$$

so the bullet does reach its target.

(b) Using Einstein's velocity addition rule, the result is instead

$$v_{\text{bullet}} = \frac{v_{\text{car}} + v_{\text{muzzle}}}{1 + \frac{v_{\text{car}} v_{\text{muzzle}}}{c^2}} = \frac{\frac{5}{6}c}{1 + \frac{1}{6}} = \frac{5}{7}c < \frac{3}{4}c$$

so the bullet does not reach its target. □

Problem 12.5

Synchronized clocks are stationed at regular intervals, a million km apart, along a straight line. When the clock next to you reads 12 noon:

(a) What time do you see on the 90th clock down the line?

(b) What time do you observe on that clock?

Solution:

(a) The 90th clock down the line is a distance $d = 90 \times 10^6$ km $= 90 \times 10^9$ m. The time it takes for light to travel this distance is

$$t = \frac{d}{c} = \frac{90 \times 10^9 \text{ m}}{3 \times 10^8 \text{ m/s}} = 300 \text{ s}.$$

This means that what you see on the 90th clock is what it showed 300 seconds ago, so the clock reads 11:55.

(b) The time you **observe** on that clock is taking into account the time it takes for the information to travel. Therefore, what you observe is what the clock really is, 12:00. □

Problem 12.6

Every 2 years, more or less, The New York Times publishes an article in which some astronomer claims to have found an object traveling faster than the speed of light. Many of these reports result from a failure to distinguish what is seen from what is observed — that is, from a failure to account for light travel time. Here's an example: A star is traveling with speed v at an angle θ to the line of sight. What is its apparent speed across the sky? (Suppose the light signal from b reaches the earth at a time Δt after the signal from a, and the star has meanwhile advanced a distance Δs across the celestial sphere; by "apparent speed," I mean $\frac{\Delta s}{\Delta t}$.) What angle θ gives the maximum apparent speed? Show that the apparent speed can be much greater than c, even if v itself is less than c.

Problem 12.7

In a laboratory experiment, a muon is observed to travel 800 m before disintegrating. A graduate student looks up the lifetime of a muon (2×10^{-6}s) and concludes that its speed was

$$v = \frac{800 \text{ m}}{2 \times 10^{-6}\text{s}} = 4 \times 10^8 \text{ m/s}.$$

Faster than light! Identify the student's error, and find the actual speed of this muon.

Solution:

The student did not take into account the lifetime of the muon in the muons reference time. The muons lifetime in its own reference time is

$$\gamma t = \frac{1}{\sqrt{1-\left(\frac{v}{c}\right)^2}}t$$

If it travels for 800m then its speed is

$$\frac{800 \text{ m}}{\gamma \times 2 \times 10^{-6}\text{s}} = v$$

Simplifying the distance over time gives

$$\frac{800 \text{ m}}{2 \times 10^{-6}\text{s}} = \frac{4}{3}c$$

Solving for v

$$\frac{4}{3}c = \gamma v = \frac{v}{\sqrt{1-\left(\frac{v}{c}\right)^2}}$$

$$\downarrow$$

$$\left(\frac{4}{3}c\right)^2 = \frac{v^2}{1-\left(\frac{v}{c}\right)^2}$$

$$\downarrow$$

$$\left(\frac{4}{3}c\right)^2\left(1-\left(\frac{v}{c}\right)^2\right) = v^2$$

$$\downarrow$$

$$\left(\frac{4}{3}c\right)^2 = v^2\left(1+\left(\frac{4}{3}c\right)^2\frac{1}{c^2}\right) = v^2\left(1+\frac{16}{9}\right)$$

$$\downarrow$$

$$v = \frac{4}{3}c\frac{1}{\sqrt{1+\frac{16}{9}}} = \frac{4}{3}c\frac{1}{\sqrt{\frac{25}{9}}} = \frac{12}{15}c = \boxed{\frac{4}{5}c}$$

□

Problem 12.8

A rocket ship leaves earth at a speed of $\frac{3}{5}c$. When a clock on the rocket says 1 hour has elapsed, the rocket ship sends a light signal back to earth.

(a) According to earth clocks, when was the signal sent?

(b) According to earth clocks, how long after the rocket left did the signal arrive back on earth?

(c) According to the rocket observer, how long after the rocket left did the signal arrive back on earth?

Solution:

(a) The time it takes for one hour to pass on the rocket, $t_r = 1\text{h}$, is according to earth the time t_e

$$t_e = \gamma t_r = \frac{1}{\sqrt{1-\frac{3^2}{5^2}}} t_r = \sqrt{\frac{25}{16}} t_r = \frac{5}{4} t_r = \frac{5}{4}\text{h}$$

(b) The time the signal from the rocket was sent, according to earth, was $\frac{5}{4}$h. This means the rocket has travelled a distance $d = t_e/(\frac{3}{5}c) = 4500\text{s} \times \frac{3}{5}c = 8.1 \times 10^{10}$ m. This distance takes 2700s to reach earth, with a total time of $4500 + 2700$ s $= 7200$ s $= 2$h

(c) We can use the result in (b), where the time on earth was 2 hours. According to the rocket, earth's clock run slow, so the time according to the rocket is

$$\gamma \times 2\text{h} = \frac{5}{4} \times 2\text{h} = 2.5\text{h}$$

Another solution, if the result for (b) is missing. Imagine that the rocket is now stationary and earth is moving away at $\frac{3}{5}c$. The signal is sent after 1 hour, and in this time the earth has travelled $60\text{min} \times \frac{3}{5}c = 6.48 \times 10^9$ m. This distance now needs to be covered by the signal, which takes $6.48 \times 10^9\text{m} \times \frac{1}{c} = 36\text{min}$. Repeating the argument, in those 36 minutes, the earth has travelled another distance, so we get $36\text{min} \times \frac{3}{5}c$. We can model this as an infinite sum, with the factor 3/5:

$$t = 60\text{min} + 36\text{min} \times \sum_0^\infty \left(\frac{3}{5}\right)^n$$

which is a geometric series with solution

$$60\text{min} + 36\text{min}\,\frac{1}{1-\frac{3}{5}} = 60\text{min} + 90\text{min} = 150\text{min} = 2\text{h}30\text{min}$$

□

Problem 12.9

A Lincoln Continental is twice as long as a VW Beetle, when they are at rest. As the Continental overtakes the VW, going through a speed trap, a (stationary) policeman observes that they both have the same length. The VW is going at half the speed of light. How fast is the Lincoln going?

Solution:

The length of the cars while stationary is denoted L. With length contraction due to the velocity accounted for the length is denoted L'. With v_{vw} given and $L_{\text{lc}} = 2L_{\text{vw}}$, the solution is obtained by setting $L'_{\text{lc}} = L'_{\text{vw}}$ and solving for v_{lc}.

$$L_{\text{lc}} = 2L_{\text{vw}}$$

$$L'_{\text{vw}} = L_{\text{vw}}\sqrt{1-\frac{v_{\text{vw}}^2}{c^2}} = L_{\text{vw}}\frac{\sqrt{3}}{2}$$

$$L'_{\text{vw}} = L'_{\text{lc}} = L_{\text{lc}}\sqrt{1-\frac{v_{\text{lc}}^2}{c^2}} = L_{\text{vw}}\frac{\sqrt{3}}{2}$$

Using $L_{\text{lc}} = 2L_{\text{vw}}$ and solving for v_{lc}, the velocity for which the length of the Lincoln Continental equals the length of the VW Beetle is

$$v_{\text{lc}} = \frac{\sqrt{13}}{4}c$$

□

Problem 12.10

A sailboat is manufactured so that the mast leans at an angle $\overline{\theta}$ with respect to the deck. An observer standing on a dock sees the boat go by at speed v. What angle does this observer say the mast makes?

Solution:

The direction of the speed is Lorentz contracted by $\frac{1}{\gamma}$. The horizontal and vertical components of the mast is

$$\overline{x} = \overline{l}\cos\left(\overline{\theta}\right), \quad \overline{y} = \overline{l}\sin\left(\overline{\theta}\right)$$

Transforming these components with regard to the velocity gives

$$x = l\frac{\cos(\theta)}{\gamma}, \quad y = l\sin(\theta)$$

This gives

$$\frac{\overline{y}}{\overline{x}} = \frac{y}{x}$$

$$\downarrow$$

$$\boxed{\tan\left(\overline{\theta}\right) = \gamma\tan(\theta)}$$

□

Problem 12.11

Problem 12.12

Solve

$$\text{(i) } \overline{x} = \gamma(x - vt)$$
$$\text{(ii) } \overline{y} = y$$
$$\text{(iii) } \overline{z} = z$$
$$\text{(iv) } \overline{t} = \gamma\left(t - \frac{v}{c^2}x\right)$$

for x, y, z, t in terms of $\overline{x}, \overline{y}, \overline{z}, \overline{t}$, and check that you recover

$$\text{(i') } x = \gamma(\overline{x} + v\overline{t})$$
$$\text{(ii') } y = \overline{y}$$
$$\text{(iii') } z = \overline{z}$$
$$\text{(iv') } t = \gamma\left(\overline{t} + \frac{v}{c^2}\overline{x}\right)$$

Solution:

From (i) and (iv) we have

$$\text{(i)} \quad \overline{x} = \gamma(x - vt) \to x = \frac{\overline{x}}{\gamma} + vt$$
$$\text{(iv) } \overline{t} = \gamma\left(t - \frac{v}{c^2}x\right) \to t = \frac{\overline{t}}{\gamma} + \frac{v}{c^2}x$$

Inserting t from (iv) into the expression for x in (i) gives

$$x = \frac{\overline{x}}{\gamma} + v\left(\frac{\overline{t}}{\gamma} + \frac{v}{c^2}x\right) \to x\left(1 - \frac{v^2}{c^2}\right) = \frac{\overline{x}}{\gamma} + \frac{v\overline{t}}{\gamma}$$
$$\downarrow$$
$$x\frac{1}{\gamma^2} = \frac{1}{\gamma}(\overline{x} + v\overline{t}) \to \boxed{x = \gamma(\overline{x} + v\overline{t}) = \text{(i')}}$$

Putting this expression for x into the expression for t in (iv) gives

$$t = \frac{\bar{t}}{\gamma} + \frac{v}{c^2}(\gamma(\bar{x} + v\bar{t})) = \gamma\left(\frac{\bar{t}}{\gamma^2} + \frac{v}{c^2}\bar{x} + \frac{v^2}{c^2}\bar{t}\right) = \gamma\left(\left(1 - \frac{v^2}{c^2} + \frac{v^2}{c^2}\right)\bar{t} + \frac{v}{c^2}\bar{x}\right)$$

$$\downarrow$$

$$\boxed{t = \gamma\left(\bar{t} + \frac{v}{c^2}\bar{x}\right) = \text{(iv')}}$$

(ii) $\longleftrightarrow$ (ii') and (iii) $\longleftrightarrow$ (iii') follow directly from the equations.

□

Problem 12.13

Sophie Zabar, clairvoyante, cried out in pain at precisely the instant her twin brother, 500 km away, hit his thumb with a hammer. A skeptical scientist observed both events (brother's accident, Sophie's cry) from an airplane traveling at $\frac{12}{13}c$ to the right. Which event occurred first, according to the scientist? How much earlier was it, in seconds?

Solution:

Using the Lorentz transformations

$$\bar{x} = \gamma(x - vt)$$

$$\bar{t} = \gamma\left(t - \frac{v}{c^2}x\right)$$

and setting Sophie's position to $x = 500\text{km}$, and her brothers position to $x = 0\text{km}$. The time of the brothers event according to the scientist is

$$\bar{t}_1 = \frac{1}{\sqrt{1 - \frac{12^2}{13^2}}}\left(0 - \frac{12}{13c}0\right) = 0\text{s}$$

The time of Sophie's event is

$$\bar{t}_2 = \frac{1}{\sqrt{1 - \frac{12^2}{13^2}}}\left(0 - \frac{12}{13c}500\text{ km}\right) = \frac{1}{\sqrt{\frac{25}{169}}}\left(-\frac{12}{13c}500\text{km}\right) =$$

$$= -\frac{13}{5}\frac{12}{13c}500\text{km} = -\frac{1200000}{c}\text{ m} = -0.004\text{ s} = -4\text{ms}$$

The twin brothers event happened 4ms earlier according to the scientist. □

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